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BRAZILIAN PHYSICAL SOCIETY

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Entangled Interlocked Diamond-Like (Diamondiynes) Lattices

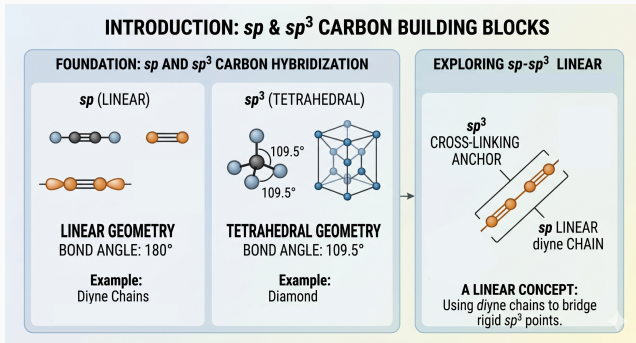
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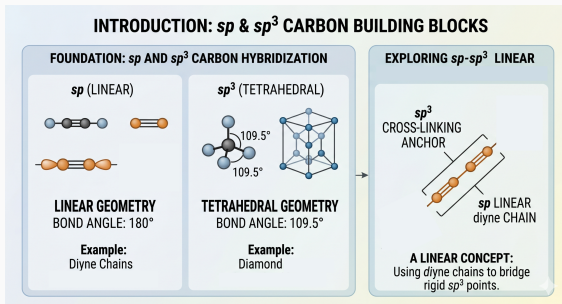
The Versatility of Carbon Allotropes

- Carbon is known for creating a wide range of distinct materials, called allotropes, mainly due to its different hybridization states: sp , sp^2 , and sp^3 .
- These hybridizations allow for materials with completely contrasting properties, ranging from soft (graphite) to ultra-hard (diamond).



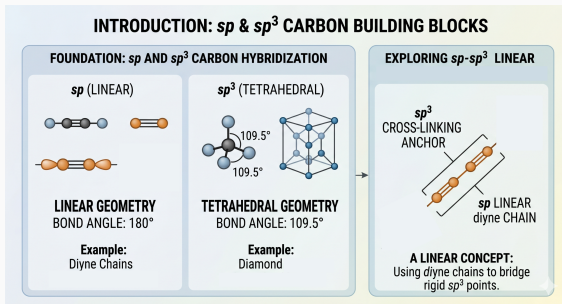
Motivation: From Linear Geometries to 3D Structures

- The sp have a **triple** ($C \equiv C$) that result a **linear** geometry
- While the standard sp^3 have a **single** ($C - C$) with tetrahedral geometry .
- These chains are formed by alternating sp and sp^3 , forming a linear carbon structure.



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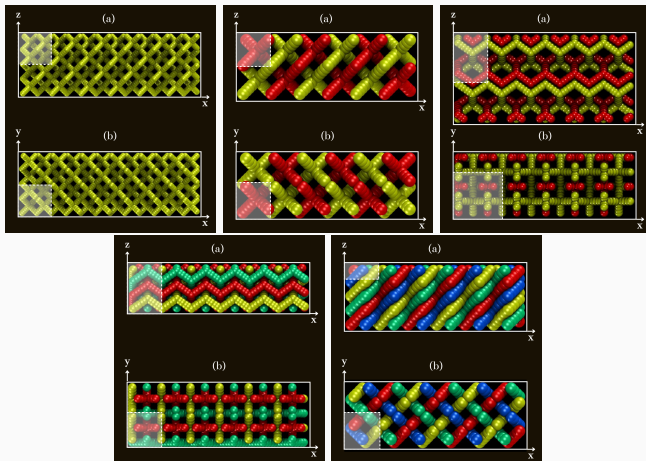
3D Architecture Formation

These linear bond sequences can be “folded” and interconnected across the tetrahedral nodes to form the independent sub-lattices that compose the entangled 3D structure.

Results: The Diamondiynes Family

A New Class: Diamondiynes

- We propose a new family of 3D carbon allotropes characterized by entangled and interlocked sublattices: unfold, two fold symmetric (2f-sym), two fold unsymmetric (2f-unsym), three fold (3f) and four fold (4f).



Bastos, C. M. de O., dos Santos, E. J. A., Alves, R. A. F., Cavalheiro Dias, A., Ribeiro Junior, L. A., Galvão, D. S. (2025). Entangled Interlocked Diamond-like (Diamondiynes) Lattices. *ACS Omega*, 10(39), 46065–46070.

We investigate the properties using First-Principles Calculations

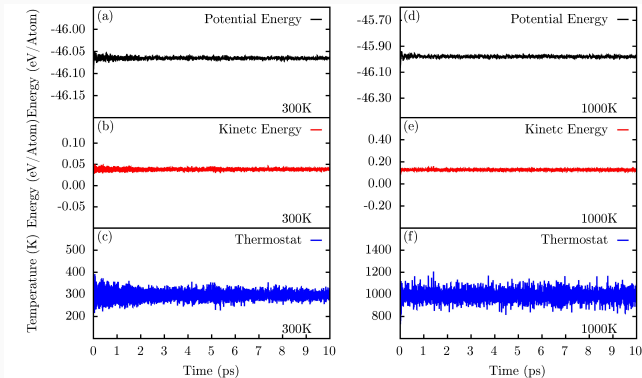
- **Density Functional Theory (DFT):**
 - Semilocal and hybrid exchange-correlation functionals to accurately describe electronic properties.
 - DFT+D3 to include the Van der Waals correction.
 - Codes utilized: **SIESTA**.
- **Ab Initio Molecular Dynamics (AIMD):**
 - Employed to evaluate the thermodynamic stability of the new structures at different temperature regimes (300 K and 1000 K).



But, this material is...

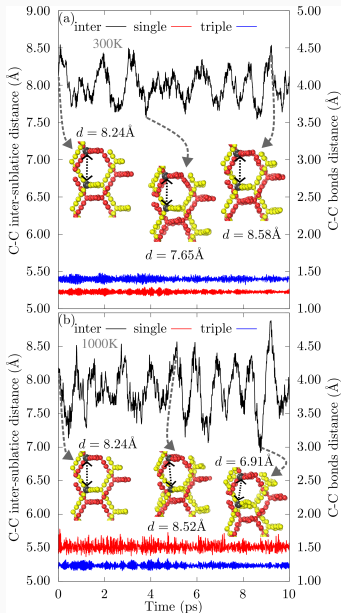
But, this material is... Moveable!

- AIMD simulations demonstrate that all five diamondyne structures are **thermodynamically stable** at 300 K and 1000 K.



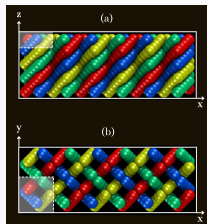
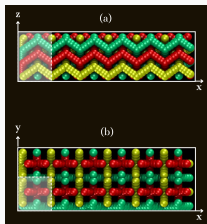
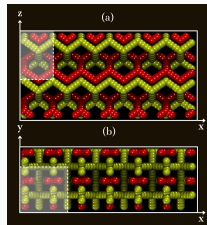
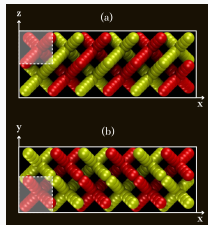
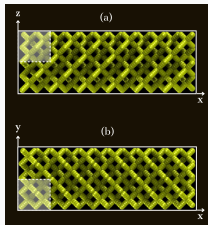
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Dynamics and Internal Mobility



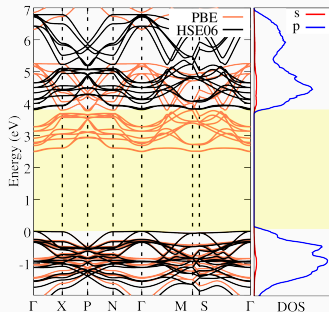
- C-C bonds within each sublattice (single and triple) remain stable even at 1000 K.
- The distance between atoms on different sublattices shows large, random fluctuations.
- **Conclusion:** This confirms the high relative mobility and independence of the interlocked sublattices.

A Unique Architecture: Interlocked and Movable Lattices



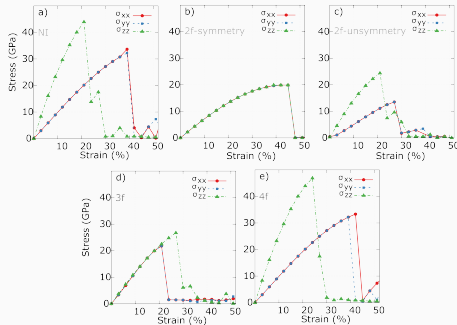
Electronic Band Gaps

- All structures are semiconductors with wide electronic band gaps (ranging from 2.2 to 4.0 eV).
- Promising for optoelectronic applications.



Mechanical Response

- Highly resistant yet flexible compared to traditional diamond-like structures.
- Potential for impact-absorbing materials.





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Article

Entangled Interlocked Diamond-like (Diamondynes) Lattices

Carlos Maciel de Oliveira Bastos, Emanuel José Alexandrino dos Santos, Rodrigo Alkimim Faria Alves, Alexandre Cavalheiro Dias, Luiz Antônio Ribeiro Junior,* and Douglas Soares Galvão

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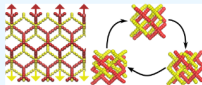
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ABSTRACT: Diamondynes, a new class of diamond-like carbon allotropes composed of carbon with sp^3/sp^2 -hybridized carbon networks, exhibit unique structural motifs that have not been previously reported in carbon materials. These architectures feature sublattices that are both interlocked and capable of relative movement. Using the density functional theory approach with semilocal and hybrid exchange-correlation functionals, we have conducted an extensive investigation into the structural and electronic properties of five diamondyne structures. Our results show that diamondynes are thermodynamically stable and exhibit wide electronic band gaps, ranging from 2.2 to 4.0 eV. They are flexible yet highly resistant compared to other diamond-like structures. They have relatively small cohesive energy values, consistent with the fact that one diamondyne structure (2f-unym) has already been experimentally realized. Our results provide new physical insights into diamond-like carbon networks and suggest promising directions for the development of porous, tunable frameworks with potential applications in energy storage and conversion.



Diamondyne: A 3D carbon allotrope with mixed valence hybridization

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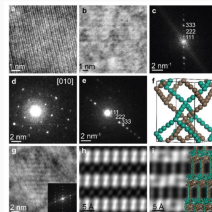
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- **A Novel Carbon Architecture:** We unveiled "Diamondiynes", a new family of 3D carbon allotropes composed of sp - sp^3 hybridized networks.
- **Unique Movability:** These structures feature a unique physical mechanism where sublattices are deeply interlocked but retain high relative mobility without breaking chemical bonds.
- **Robust Stability:** AIMD simulations confirm thermodynamic stability at high temperatures (up to 1000 K).
- **Broad Application Potential:** With wide band gaps (2.2 – 4.0 eV) and a combination of mechanical flexibility and strength, these tunable frameworks open new pathways for energy storage, conversion, and impact-absorption technologies.



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We're always open to new collaborations...

...especially if they involve discussing
physics over a good cup of coffee!



Thank you!

Our Group: LCCMat - IF/UnB



Funding Agencies:

